

bargh-et-al-2015-automaticity-of-social-behavior--direct-e

Auto-assembled report. Section content is drawn from the summarization pipeline outputs.

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Paper Synthesis

Paper:

bargh-et-al-2015-automaticity-of-social-behavior--direct-effects-of-trait-construct-and-stereotype-activation-on-a

Executive Summary

The paper investigates whether priming trait concepts and stereotypes can directly trigger corresponding social behaviors without conscious mediation. Across three experiments, priming rudeness led participants to interrupt an experimenter more quickly, elderly-stereotype priming slowed walking speed, and African-American stereotype priming increased observed hostility, supporting the claim of automatic behavior activation.

Reading Recommendation

Read Selectively

The paper offers a novel claim that primed traits and stereotypes can directly trigger behavior, supported by statistically significant effects in Experiments 1 and 2, but methodological weaknesses (e.g., conflated outcome in Exp 1, limited behavioral measures, missing statistics for Exp 3, no power analyses) reduce confidence in the central theory. Scholars should focus on the introduction, the detailed methods and results of Experiments 1 and 2, and the general discussion to assess the theoretical contribution and the robustness of the evidence.

Background and Motivation

The authors identify a specific gap: it is unclear whether the passive, automatic effects of priming extend beyond social perception to directly shape behavioral responses. This question builds on Bargh's (1994) review of automatic activation of trait constructs, Higgins's (1989) demonstration of priming influences on impression formation, and Wyer & Srull's (1989) evidence for unintended, passive priming effects. By targeting this gap, the study seeks to test whether trait and stereotype priming can produce immediate, non-conscious changes in overt behavior, extending the prior literature that has focused largely on perceptual and attitudinal outcomes.

Theoretical Stance

Overall stance: Consistent Theoretical

The introduction foregrounds a focal theory that primed trait constructs and stereotypes can directly activate corresponding behaviors, and the discussion repeatedly returns to this claim while integrating related perspectives (e.g., perception-behavior link) and rejecting mediational accounts, showing alignment across sections.

Theories Discussed

- **automatic activation of behavioral responses** (*focal*): The paper's central claim that trait concepts and stereotypes, once primed, can become automatically activated and directly trigger corresponding social behaviors without conscious mediation.
- **mediational model of behavior** (*alternative*): A competing account that posits social perceptions and judgments mediate the effect of priming on behavior; the authors argue against this model.
- **situationally linked behavior** (*background*): The view (Lewin, 1943; Mischel, 1973; Berkowitz, 1984; Higgins, 1987) that individuals possess behavioral responses that are associatively linked to specific situations.

- **automatic perception-behavior link** (*alternative*): The perspective (Berkowitz, 1984; Carver et al., 1983) that perception and behavior can be automatically coupled, consistent with the paper's findings.
- **behavioral schema model** (*background*): Carver et al.'s (1983) model proposing that activated behavioral schemas guide action, cited as harmonious with the present results.
- **pre-existing behavioral representations** (*background*): The notion that automatic social behavior occurs only when the relevant behavioral representation is already linked to the situation for the individual.
- **control of automatic influences** (*background*): Bargh (1989) and Wegner (1994) framework outlining awareness, motivation, and attentional capacity as prerequisites for controlling automatic effects.
- **subliminal advertising debate** (*background*): The broader controversy over whether subliminal messages can automatically induce consumer behavior; mentioned to contrast with the present priming effects.

Study Design

- **Number of experiments:** 3
- **Predictors examined:** priming (rude vs. polite vs. neutral), prime condition (elderly vs. neutral), priming (African American face vs. Caucasian face)
- **Outcomes examined:** interruption waiting time (seconds until participant interrupts experimenter), walking time (seconds to walk 9.75 m corridor), hostility rating (0-10 scale coded from videotaped facial expressions)
- **Materials:** Experiment 1 used a scrambled-sentence task (30 items, 15 critical words per version) to prime trait concepts. Experiment 2 employed a similar scrambled-sentence task with elderly-related words and measured walking speed with a hidden stopwatch. Experiment 3 presented subliminal face primes (13-26 ms) masked by pattern masks, followed by a target display, and coded facial hostility from video recordings.
- **Design types:**
 - exp_1: between-subjects 3-level priming (rude, polite, neutral)
 - exp_2: 2 × 2 between-subjects design (elderly vs. neutral prime × experiment 2a vs. 2b)
 - exp_3: between-subjects 2-level priming (African American vs. Caucasian face)

Findings Summary

Across the three experiments, priming a trait concept or stereotype led to behavior that matched the content of the prime in a subsequent, unrelated context: rude-concept priming accelerated participants' interruptions of an experimenter, elderly-stereotype priming slowed walking speed, and African-American face priming increased observer-rated hostility after a task error. The effects were observed despite participants reporting no differences in perceived politeness of the experimenter and despite the authors' claim that mood did not mediate the walking-speed outcome. However, the magnitude and reliability of the effects varied, with Experiment 1 showing a ceiling-effect in interruption times, Experiment 2 showing consistent slowing of walking speed across two replications but no reported interaction effects, and Experiment 3 lacking reported statistical tests for the hostility difference. Together, the results support the authors' hypothesis that activated trait or stereotype representations can directly trigger corresponding social behavior, while also highlighting methodological limitations that temper confidence in the uniformity of the automatic behavior link.

Evidence-Claim Alignment

Supported Claims

- The authors claim that primed trait concepts can directly trigger corresponding social behavior; Experiment 1 provides statistical support ($F(2,33)=5.76$, $p=.008$) showing that participants primed with rudeness interrupted the experimenter faster than those primed with politeness or neutral stimuli.
- The claim that activating an elderly stereotype leads to slower walking speed is supported by Experiment 2a ($t(28)=2.86$, $p<.01$, $M_{\text{elderly}}=8.28$ s vs $M_{\text{neutral}}=7.30$ s) and Experiment 2b ($t(28)=2.16$, $p<.05$, $M_{\text{elderly}}=8.20$ s vs $M_{\text{neutral}}=7.23$ s).

Unsupported or Weakly Supported Claims

- The assertion that the African-American stereotype priming produces greater hostility is not substantiated in the record; no statistical test or effect size for the hostility rating is reported in Experiment 3, leaving the claim unsupported.
- The claim that the interruption latency in Experiment 1 reflects a direct, non-mediated behavioral effect is weakened by the authors' own discussion of an alternative perceptual explanation and by the critique noting that the measure conflates participants' perception of the experimenter's rudeness with their own behavior.
- The statement that the elderly-stereotype effect on walking speed is purely nonconscious is undermined by reported mood differences (more positive mood in the elderly-prime condition) and the critique's note that mood mediation was not formally tested, making the claim only weakly supported.

Contradictions or Tensions

- Tension between the theoretical claim of a direct automatic behavior link and the methodological concern that Experiment 1's outcome (interruption waiting time) may be driven by participants' appraisal of the experimenter's behavior rather than a pure trait-behavior enactment.
- Conflict between the authors' dismissal of mood as a mediator in Experiment 2 and the evidence that the elderly-prime condition produced a significantly more positive mood, coupled with the absence of mediation analysis, creating a tension between the claimed nonconscious effect and a plausible affective pathway.
- The general discussion extrapolates the findings to a broad automatic perception-behavior linkage across diverse contexts, yet the critique and authors' own limitations highlight that each experiment employed highly constrained behavioral measures, generating a contradiction between the scope of the theoretical conclusion and the narrow empirical base.

Strengths

- Demonstrates converging evidence across three distinct experiments that priming trait concepts or stereotypes can influence overt behavior in unrelated contexts.
- Employs a variety of behavioral measures (interruption latency, walking speed, facial hostility) to test the automatic behavior hypothesis, enhancing construct validity.
- Uses well-established priming paradigms (scrambled-sentence tasks) and extends them to novel outcome domains, showing methodological creativity.
- Provides clear statistical support for the primary effects in Experiments 1 ($F(2,33)=5.76$, $p=.008$) and 2 ($t(28)=2.86$, $p<.01$; $t(28)=2.16$, $p<.05$), indicating reliable priming-behavior links within the sampled populations.
- Integrates the findings with multiple theoretical perspectives (situationally linked behavior, automatic perception-behavior link, behavioral schema model), offering a coherent theoretical narrative.

Weaknesses and Concerns

- Experiment 1's primary outcome (interruption waiting time) conflates participants' own behavior with their perception of the experimenter's rudeness, weakening the claim of a direct trait-behavior link and introducing a plausible mediated explanation.
- The priming manipulations contain substantial filler content and, in Experiment 2, capture only a single facet (walking speed) of the multifaceted elderly stereotype, limiting fidelity to

the theoretical construct being tested.

- Key statistical reporting is incomplete: Experiment 3 provides no test statistics for the hypothesized hostility difference, and Experiment 2 fails to report the interaction or main effect of the experiment factor in the 2×2 design, impeding assessment of robustness.
- Methodological reporting gaps--absence of power analyses, exclusion criteria, counterbalancing details, and reliability metrics (e.g., inter-rater reliability for hostility coding)--reduce confidence in the adequacy and reproducibility of the designs.
- Potential alternative mediators (e.g., mood differences in Experiment 2) are noted but not formally tested, and the authors dismiss them without mediation analyses, leaving the non-conscious interpretation under-supported.

Key References

- **Bargh (1994)** -- foundational theory
- **Higgins (1989)** -- foundational theory
- **Wyer & Srull (1989)** -- foundational theory
- **Srull & Wyer (1979)** -- primary methodological precedent
- **Devine (1989)** -- key empirical precedent
- **Carver et al. (1983)** -- competing theoretical account
- **Berkowitz (1984)** -- competing theoretical account
- **Lewin (1943)** -- foundational theory
- **Mischel (1973)** -- foundational theory
- **Bargh (1989)** -- primary methodological precedent
- **Wegner (1994)** -- primary methodological precedent
- **Bem (1972)** -- competing theoretical account
- **Bargh (in press)** -- primary methodological precedent
- **Kihlstrom, Barnhardt, & Tatarzyn (1992)** -- key empirical precedent
- **Moore (1982)** -- key empirical precedent

Abstract

Preserved from source without summarization.

Previous research has shown that trait concepts and stereotypes become active automatically in the presence of relevant behavior or stereotyped-group features. Through the use of the

same priming procedures as in previous impression formation research, Experiment 1 showed that participants whose concept of rudeness was primed interrupted the experimenter more quickly and frequently than did participants primed with polite-related stimuli. In Experiment 2, participants for whom an elderly stereotype was primed walked more slowly down the hallway when leaving the experiment than did control participants, consistent with the content of that stereotype. In Experiment 3, participants for whom the African American stereotype was primed subliminally reacted with more hostility to a vexatious request of the experimenter. Implications of this automatic behavior priming effect for self-fulfilling prophecies are discussed, as is whether social behavior is necessarily mediated by conscious choice processes.

Introduction (Paper-Level)

Section: paper-level

Research Gap / Motivation

It remains unclear whether the passive, automatic effects of priming extend beyond social perception to directly influence behavioral responses.

Theoretical Framework

automatic priming effects on behavior

We argue here that such passive, automatic effects of priming need not be limited to social perception.

mental representation activation

assuming that behavioral responses to situations are also represented mentally, as are stereotypes and attitudes, they should also be capable of becoming automatically activated, by the same principles that govern the development of automaticity of other representations.

Mechanistic Claims

automatic activation of behavioral responses

assuming that behavioral responses to situations are also represented mentally, as are stereotypes and attitudes, they should also be capable of becoming automatically activated, by the same principles that govern the development of automaticity of other representations.

Constructs of Interest

- **priming (activation of trait constructs or stereotypes)** (*predictor*): incidental activation of knowledge structures by the current situational context
- **behavioral responses** (*outcome*): actions or responses to situations that are represented mentally and can be automatically activated

Rationale for Predictor

Priming has been shown to unintentionally influence social perception, suggesting it can serve as a powerful, automatic cue that shapes subsequent mental processing.

Rationale for Outcome

If behavioral responses are mentally represented like stereotypes and attitudes, they should be susceptible to the same automatic activation processes.

Key Terminology

Priming: the incidental activation of knowledge structures, such as trait constructs and stereotypes, by the current situational context

Priming refers to the incidental activation of knowledge structures, such as trait concepts and stereotypes, by the current situational context.

Background Research

- Bargh (1994) reviewed numerous studies showing that recent use of a trait construct or stereotype can unintentionally influence interpretation of behavior.
- Higgins (1989) documented automatic influences of primed constructs on social perception.
- Wyer & Srull (1989) provided evidence for unintended, passive effects of priming on impression formation.

Missing or Unclear

- specific hypotheses
- alternative theories
- detailed overview of planned experiments
- empirical predictions

All Citations Referenced

- **Bargh (1994)** -- review of priming effects on social perception and automaticity
- **Higgins (1989)** -- evidence for automatic priming influences on impression formation
- **Wyer & Srull (1989)** -- review of passive, unintended priming effects

Additional Quoted Evidence

Priming refers to the incidental activation of knowledge structures, such as trait concepts and stereotypes, by the current situational context.

We argue here that such passive, automatic effects of priming need not be limited to social perception.

assuming that behavioral responses to situations are also represented mentally, as are stereotypes and attitudes, they should also be capable of becoming automatically activated, by the same principles that govern the development of automaticity of other representations.

Experiment 1 -- Introduction

Not available for this paper.

Experiment 1 -- Methods

Section: exp_1

Participants

- **N:** 34 students
- **Recruitment:** Introductory Psychology course at New York University
- **Testing context:** Individual lab sessions; participants tested one at a time in a laboratory waiting room

Design

Design statement: 3 (priming: rude vs. polite vs. neutral) between-subjects design.

Predictors:

- **priming**
- Levels: rude, polite, neutral
- Manipulation: between-subjects
- Factor type: nominal

Counterbalancing: not reported

Outcome Measures

- **interruption waiting time:** elapsed time recorded by confederate stopwatch from when participant entered the hallway until participant spoke to experimenter; maximum 10 minutes

Materials

- **Description:** Scrambled Sentence Test consisting of 30 items; participants construct a four-word sentence from five scrambled words as quickly as possible
- **Source:** adapted from Srull & Wyer (1979)
- **Stimulus pool size:** 30 items
- **Stimuli presented per participant:** 30 items per participant
- **Additional details:**
 - Three versions (rude, polite, neutral) differing in 15 critical words per version
 - Critical words listed for each version in the text

Procedure

Ordered steps:

1. Participant arrives at central waiting area and is greeted by experimenter
2. Consent obtained
3. Experimenter explains two short studies about language ability
4. Participant receives envelope with one version of scrambled-sentence test
5. Participant completes the 30-item test (≈ 5 min)
6. Participant exits to hallway and approaches second room where experimenter and confederate are conversing
7. Experimenter gives subtle prearranged sign; confederate starts stopwatch

8. Participant waits; stopwatch records time until participant interrupts with a phrase such as "Excuse me"
 9. If no interruption within 10 min, stopwatch stops at 10 min
 10. Participant completes brief anagram puzzle (≤ 2 min)
 11. Partial debrief, then survey administered by second confederate
 12. Full debrief and participant leaves
- **Experimenter presence:** present in the laboratory rooms
 - **Presentation format:** not reported
 - **Stimulus timing:** scrambled-sentence test self-paced; interruption waiting time self-paced up to 10 min

Preregistration and Analysis Plan

- **Preregistration:** not reported

Missing or Unclear

- Compensation not specified
- Participant demographics (age, gender, etc.) not reported
- Power analysis not reported
- Exclusion criteria not reported
- Apparatus/software details not reported
- Counterbalancing scheme not reported
- Exact stimulus timing parameters for the waiting period not detailed beyond max 10 min
- Preregistration status not reported
- Data analysis plan not reported
- Total session duration not reported

Citations Referenced in This Section

- **Srull & Wyer (1979)** -- source of the Scrambled Sentence Test used as the priming manipulation

Quoted Evidence from Source

A total of 34 students at New York University who were enrolled in the Introductory Psychology course participated in the experiment

The priming manipulation took the form of a "Scrambled Sentence Test" (Srull & Wyer, 1979)

Our dependent measure was the amount of time the participant would wait until interrupting the conversation

Experiment 1 -- Results and Discussion

Section: exp_1

Statistical Approach

A one-way analysis of variance (ANOVA) of these data, with priming condition as the single factor, revealed a significant main effect. We applied the test for a linear trend in proportions. Because we made a specific directional prediction regarding the ordering of the three percentages, rather than an omnibus prediction of any difference among the three, we applied the test for a linear trend in proportions.

Key Findings

Participants in the rude priming condition interrupted significantly faster than those in the neutral or polite conditions

- **Test:** one-way ANOVA
- **Effect type:** main effect
- **Variables in model:** priming condition, time to interruption

Verbatim statistical report:

$F(2,33) = 5.76, p = .008$

Verbatim descriptives:

Participants in the rude priming condition interrupted significantly faster ($M = 326$ s) than did participants in the neutral ($M = 519$ s) or polite ($M = 558$ s) priming conditions

The rude prime condition mean was significantly shorter than each of the other two means

- **Test:** simple t tests

- **Effect type:** planned comparison
- **Variables in model:** priming condition, time to interruption

Verbatim statistical report:

| both ps < .04

The proportion of participants who interrupted the experimenter increased as a function of the trait concept primed, from politeness through no priming to rudeness

- **Test:** test for a linear trend in proportions
- **Effect type:** linear trend
- **Variables in model:** priming condition, percentage of participants who interrupted

Verbatim statistical report:

| Z = 2.32, p = .02, two-tailed

- Citations: Snedecor & Cochran (1980)

There was no reliable difference in the ratings of the experimenter's politeness across the three priming conditions

- **Test:** one-way ANOVA
- **Effect type:** main effect
- **Variables in model:** priming condition, experimenter's politeness ratings

Verbatim statistical report:

| F(2,28) < 1

Verbatim descriptives:

| polite prime condition: M = 0.6, rude prime condition: M = 0.4, neutral prime condition: M = 0.8

Authors' Interpretations

direct effect on behavior

| Whereas our interpretation of this result is that the priming manipulation produced a direct, preconscious effect on the participants' subsequent behavior in a social

interaction, an alternative explanation in terms of the social perceptual effects of priming must be considered.

automatic activation of trait concepts

The results of Experiment 1 showed that the passive, automatic activation of a trait concept results in traitlike behavior by the individual, but there are two ways in which trait concepts can be activated directly by the environment.

stereotype activation

The deduction that follows naturally from these two sets of findings is that the priming or automatic activation of stereotypes should make the perceiver him or herself more likely to act in accordance with the trait concepts that participate in that stereotype.

Authors' Stated Links to Their Own Predictions

support for hypothesis

Although this result supports our hypothesis that social interaction behavior can be primed, the time-to-interruption distribution varied considerably from normality.

support for linear trend hypothesis

In support of the hypothesis, the test revealed a reliable linear trend ($Z = 2.32$, $p = .02$, two-tailed).

Anomalous or Unexpected Results (as characterized by authors)

- The time-to-interruption distribution varied considerably from normality. Fully 21 of the 34 participants did not interrupt at all in the 10 min available to them, so that the time variable suffered from a severe ceiling effect.

Alternative Interpretations Discussed by Authors

- This alternative centers on how the participant may have interpreted or perceived the experimenter's behavior during his conversation with the confederate. If the experimenter's behavior was ambiguous enough to permit different impressions of him, depending on the relative accessibility of the participant's relevant trait constructs, it is possible that rude-primed participants were more likely to perceive the experimenter as rude because he was ignoring them and attending to the confederate, whereas polite-primed participants were more likely to perceive the experimenter as polite because he so patiently dealt with the confederate's questions.

Limitations Stated by Authors

- We did not design or intend the experimenter's behavior to be ambiguous in this way, because our intention was to demonstrate an effect of priming on behavior that was not mediated by differential interpretations of the experimenter's behavior.

Preview of Next Experiment

We designed Experiments 2 and 3 to test this hypothesis in the context of two distinct stereotypes: one for elderly people and one for African Americans.

Missing or Unclear (Non-Statistical)

- null

Citations Referenced

- **Snedecor & Cochran (1980)** -- pp.206-208
- **Higgins & King (1981)** -- null
- **Neuberg (1988)** -- null
- **Herr (1986)** -- null
- **Bargh & Thein (1985)** -- null
- **Higgins (1989)** -- null
- **Bargh et al. (1986)** -- null
- **Srull & Wyer (1979)** -- null
- **Newman & Uleman (1989)** -- null
- **Brewer (1988)** -- null
- **Devine (1989)** -- null
- **Fiske & Neuberg (1990)** -- null
- **Hayes-Roth (1977)** -- null
- **Bargh & Pietromonaco (1982)** -- null

Experiment 2a and 2b -- Introduction

Not available for this paper.

Experiment 2a and 2b -- Methods

Section: *exp_2a_and_2b*

Participants

- **N:** 60 (30 in Experiment 2a, 30 in Experiment 2b)
- **Recruitment:** New York University undergraduates
- **Testing context:** individual, lab, session length not reported

Design

Design statement: 2 (prime condition: elderly vs neutral) x 2 (experiment: 2a vs 2b) between-subjects design, with prime condition assigned between participants in each experiment.

Predictors:

- **prime condition**
- Levels: elderly prime, neutral prime
- Manipulation: between-subjects
- Factor type: nominal

Counterbalancing: not reported

Outcome Measures

- **walking time:** time in seconds to walk 9.75 m down the corridor measured with a hidden stopwatch

Materials

- **Description:** scrambled-sentence task with 30 sets of five word combinations; participants write a sentence using four of the five words
- **Source:** constructed by authors; prime words obtained from Brewer, Dull, & Lui (1981); Harris & Associates (1975); McTavish (1971); Perdue & Gurtman (1990)
- **Stimuli presented per participant:** 30 sets per participant
- **Additional details:**
- self-paced

- paper-and-pencil format

Procedure

Ordered steps:

1. Participant informed that the study investigates language proficiency and given scrambled-sentence task instructions
 2. Experimenter leaves the room while participant completes the self-paced task
 3. Participant notifies experimenter upon completion
 4. Experimenter re-enters and partially debriefs the participant
 5. A confederate records walking time with a hidden stopwatch as participant walks 9.75 m down the corridor
 6. Experimenter catches up with participant near the elevator and gives complete debriefing
- **Apparatus:** paper scrambled-sentence task; hidden stopwatch for walking time
 - **Experimenter presence:** present initially, absent during task, returns after task
 - **Presentation format:** not reported
 - **Stimulus timing:** self-paced

Preregistration and Analysis Plan

- **Preregistration:** not reported

Author-Provided Rationale for Methodological Choices

- to study the effect of activation of the elderly stereotype on behavior

Missing or Unclear

- power analysis not reported
- compensation details not reported
- exclusion criteria not reported
- per-cell sample size not specified
- demographic details (age, gender breakdown) not provided beyond total N
- counterbalancing scheme not described

- exact stimulus timing parameters (e.g., ISI) not reported
- session duration not reported
- apparatus details (e.g., software) not reported

Citations Referenced in This Section

- **Brewer, Dull, & Lui (1981)** -- source of elderly prime words
- **Harris & Associates (1975)** -- source of elderly prime words
- **McTavish (1971)** -- source of elderly prime words
- **Perdue & Gurtman (1990)** -- source of elderly prime words

Quoted Evidence from Source

Thirty male and female New York University undergraduates ... participated in Experiment 2a, and a different sample of 30 participated in Experiment 2b.

The participant was instructed to write down a grammatically correct sentence using only four of the five words given.

Using a hidden stopwatch, ... recorded the amount of time in seconds that the participant spent walking a length of the corridor ... 9.75 m away.

Experiment 2a and 2b -- Results and Discussion

Section: exp_2a_and_2b

Statistical Approach

null

Key Findings

Participants in the elderly priming condition had a slower walking speed compared to participants in the neutral priming condition in Experiment 2a.

- **Test:** t test
- **Effect type:** main effect
- **Variables in model:** priming condition, walking speed

Verbatim statistical report:

| $t(28) = 2.86, p < .01$

Verbatim descriptives:

| Participants in the elderly priming condition ($M = 8.28$ s) had a slower walking speed compared to participants in the neutral priming condition ($M = 7.30$ s)

Participants in the elderly priming condition had a slower walking speed compared to participants in the neutral priming condition in Experiment 2b.

- **Test:** t test
- **Effect type:** main effect
- **Variables in model:** priming condition, walking speed

Verbatim statistical report:

| $t(28) = 2.16, p < .05$

Verbatim descriptives:

| Participants in the elderly priming condition ($M = 8.20$ s) had a slower walking speed compared to participants in the neutral priming condition ($M = 7.23$ s)

The main effect of priming condition was not significant across the two dependent measures of emotion and arousal.

- **Test:** MANOVA
- **Effect type:** main effect
- **Variables in model:** priming condition, emotion, arousal

Verbatim statistical report:

| $F(1,31) < 1$

Verbatim descriptives:

| Participants in the elderly priming condition were in a more positive mood ($M = 1.7$) than control participants ($M = 0.3$) and were more aroused or energetic as well ($M_s = -0.5$ and -1.2 , respectively).

There was no interaction between priming condition and the two dependent measures of emotion and arousal.

- **Test:** MANOVA

- **Effect type:** interaction
- **Variables in model:** priming condition, emotion, arousal

Verbatim statistical report:

| $F(1,31) < 1$

Authors' Interpretations

stereotype activation influence on behavior

| The results of the present experiments suggest that exposing individuals to a series of words linked to a particular stereotype influences behavior nonconsciously. How the activated stereotype influences behavior depends on the content of the activated stereotype itself, not the stimulus words actually presented.

nonconscious effect of elderly priming

| Thus, it appears safe to conclude that the effect of the elderly priming manipulation on walking speed occurred nonconsciously.

mood mediation explanation

| The alternative explanation for our findings in terms of a mediating effect of mood caused by the elderly stereotype primes appears untenable.

Authors' Stated Links to Their Own Predictions

priming effect on walking speed

| Participants in the elderly priming condition ($M = 8.28$ s) had a slower walking speed compared to participants in the neutral priming condition ($M = 7.30$ s), $t(28) = 2.86$, $p < .01$, as predicted.

Alternative Interpretations Discussed by Authors

- One alternative explanation that can be offered for the effect of the elderly-stereotype-related stimuli on walking speed is that, in general, the words relating to the elderly stereotype are more likely than control words to induce in participants a sad mood, which might then be the reason they walked more slowly.

Citations Referenced

- **Bargh (1992)** -- review in Bargh,1992
- **Lombardi, Higgins, & Bargh (1987)** -- see Lombardi, Higgins, & Bargh,1987
- **Strack & Hannover (1996)** -- see Strack & Hannover,1996
- **Page (1969)** -- modeled after Page (1969)
- **Salovey & Birnbaum (1989)** -- modified version of the Affect-Arousal Scale (Salovey & Birnbaum,1989)

Experiment 3 -- Introduction

Section: exp_3

Rationale for This Experiment

To test whether the automatic activation effects observed for the elderly stereotype generalize to the African American stereotype and to examine hostile behavior resulting from that activation.

Relation to Prior Experiments

Experiment 3 builds on Experiment 2, which showed automatic activation of the elderly stereotype influencing behavior; this experiment assesses the same mechanism with a different stereotype.

Specific Predictions

hostile behavior prediction

According to the present model, this automatic activation should also produce a tendency toward hostile behavior in the perceiver.

- Citations: Devine (1989)

Theoretical Refinements (relative to paper-level intro)

extension to African American stereotype

Experiment 3 was intended to assess the generality of these results to an entirely different stereotype--that for African Americans.

New Elements Introduced

- Priming stimuli presented subliminally
- Different target stereotype (African American instead of elderly)

Constraints Addressed

- Ruling out experimenter demand or conscious strategic processes by using subliminal presentation

Citations Referenced

- **Devine (1989)** -- stereotype becomes active automatically to influence perceptions of a target's hostility

Experiment 3 -- Methods

Section: exp_3

Participants

- **N:** 41
- **Recruitment:** New York University
- **Compensation:** partial course credit
- **Testing context:** individual, lab, session length not reported

Design

Design statement: 1 (priming: African American vs Caucasian) between-subjects design.

Predictors:

- **priming**
- Levels: African American face, Caucasian face
- Manipulation: between-subjects
- Factor type: nominal

Counterbalancing: not reported

Outcome Measures

- **hostility rating:** 5-point unipolar scale (0-10) rated by two blind coders from videotaped facial expressions, averaged across coders

Materials

- **Description:** Black-and-white photographs of young African American or Caucasian male faces, two pattern masks, and target pictures composed of 4 to 20 colored circles on a gray background
- **Source:** in-house stimuli, pretested for masking effectiveness
- **Stimuli presented per participant:** 130 trials per participant
- **Additional details:**
 - face presentation 13-26 ms (monitor refresh 76 Hz)
 - target picture displayed for 3 s
 - masking procedure with two successive pattern masks

Procedure

Ordered steps:

1. Participant brought individually to experimental room
 2. Given written directions about odd-even judgment task
 3. Practice trials administered; experimenter leaves room
 4. Experimental trials (130) with subliminal face prime, masks, and circle target
 5. On trial 130, computer displays error message and participant is called back
 6. Experimenter announces need to repeat task, then rescinds the instruction
 7. Videotaped facial reactions captured throughout
 8. Participant completes two questionnaires (Racial Ambivalence Scale, Modern Racism Scale)
 9. Experimenter rates irritability, hostility, anger, uncooperativeness on 10-point scales
 10. Suspicion probe and debriefing
- **Apparatus:** Gateway 486 computer with VGA monitor, Visual Basic program, hidden video camera
 - **Experimenter presence:** present initially, leaves room during task, returns after error message

- **Presentation format:** trials presented sequentially; participant randomly assigned to priming condition by computer
- **Stimulus timing:** face 13-26 ms, masks similar duration, target 3 s, inter-trial interval not reported

Preregistration and Analysis Plan

- **Preregistration:** not reported

Missing or Unclear

- no power analysis reported
- exclusion criteria not stated
- per-cell N not provided
- exact stimulus timing beyond face duration not detailed (ISI, mask duration)
- session duration not reported
- gender or other demographic details absent
- details of randomization procedure beyond computer assignment not described
- compensation specifics beyond partial course credit not detailed
- number of stimuli in stimulus pool not reported

Citations Referenced in This Section

- **Katz & Hass (1988)** -- source of the Racial Ambivalence Scale used after the experiment
- **McConahay (1986)** -- source of the Modern Racism Scale used after the experiment

Quoted Evidence from Source

Participants were 41 non-African-American undergraduate students from New York University who participated in the experiment to partially fulfill course credit.

The hypothesis was that participants presented subliminally with African American faces would react with greater hostility to the computer error, compared to participants primed with Caucasian faces.

Two coders who were blind to experimental conditions and hypotheses rated all the videotaped facial expressions ... on a 5-point unipolar scale of hostility ranging from 0 (not at all hostile) to 10 (extremely hostile).

Experiment 3 -- Results and Discussion

Not available for this paper.

General Discussion

Section: paper-level

Summary of Findings

Across three experiments, trait and stereotype priming automatically activated corresponding behavioral responses in unrelated contexts, even without participants' awareness, demonstrating that behavior can be directly linked to situational representations and that such effects are stronger than impression effects under the experimental conditions.

Theoretical Conclusions

situationally linked behavior

One explanation for the present results is in terms of situationally linked behavior; that is, that a person has behavioral responses (e.g., assertiveness, anger, patience) that are associatively linked to particular situations (e.g., being made to wait by another person, losing one's work because of another's mistake).

- Citations: Lewin (1943); Mischel (1973); Berkowitz (1984); Higgins (1987)

automatic perception-behavior link

another theoretical perspective consistent with the present findings is that of an automatic perception-behavior link (e.g., Berkowitz, 1984; Carver et al., 1983).

- Citations: Berkowitz (1984); Carver et al. (1983)

behavioral schema model

Thus, our results are in harmony with those of Carver et al. (1983), who advanced a "behavioral schema" model as an explanation for modeling effects.

- Citations: Carver et al. (1983)

pre-existing behavioral representations

automatic social behavior may occur only if the behavioral representation that is activated is already associated with that situation by the individual.

control of automatic influences

Control over automatic influences requires three things: (a) awareness of the influence or at least the possibility of the influence, (b) motivation to exert the control, and (c) enough attentional capacity (or lack of distractions) at the time to engage in the control process (see Bargh, 1989; Wegner, 1994).

- Citations: Bargh (1989); Wegner (1994)

Cross-Experiment Integration

consistent priming across experiments

Across three experiments, the activation of a trait construct or a stereotype in one context resulted in behavior consistent with it in a subsequent unrelated context.

Results Authors State Support Their Theory

situationally linked behavior support

The participants were not aware of the influence or potential influence of the priming events on their behavior. The same priming techniques that have been shown in prior research to influence impression formation produce similar effects when the dependent measure is switched to social behavior.

perception-behavior link support

Because our studies used the same kind of trait and stereotype priming manipulations that have been shown by prior research to affect impression formation (e.g., Devine, 1989; Srull & Wyer, 1979), we now know the identical priming manipulations result simultaneously in automatic perceptual and behavioral effects.

- Citations: Devine (1989); Srull & Wyer (1979)

Results Authors State Challenge a Theory

challenge to mediational model

It must be noted that the evidence for such conscious mediation of behavior by attributions or judgments is historically weak... the behavioral measure showed clear predicted effects, and the evidence of the "mediator" was nonexistent.

- Citations: Bem (1972); Bargh (in press)

Alternative Accounts Addressed by Authors

mediational model of behavior (*ad-hoc alternative*)

- **Description:** The view that social perceptions and judgments mediate the effect of priming on behavior.
- **How authors addressed it:** argued against, presenting evidence that behavior can be directly triggered without mediation
- **Citations:** Bem (1972); Bargh (in press)

subliminal advertising debate (*ad-hoc alternative*)

- **Description:** The claim that subliminal messages can automatically induce consumer behavior.
- **How authors addressed it:** contrasted with their findings, explaining why similar priming did not work in movie theater contexts
- **Citations:** Kihlstrom, Barnhardt, & Tatarzyn (1992); Moore (1982)

Implications for the Field

automatic linkage of behavior to situations

The present results show that these behavioral responses become automatically linked to representations of social situations just as previous research has found perceptual trait constructs, stereotypes, and attitudes to become automatically activated.

limits of mediation models

Perceptions may play a role, but they are one of several influences, and it is clear from the present results that behaviors can be triggered directly without being mediated by impressions or judgments of the person with whom one is interacting.

Applied Implications Stated by Authors

- Automatic priming may underlie self-fulfilling prophecy effects in stereotype confirmation.
- Nonconscious activation of stereotypes can lead to hostile behaviors that reinforce the stereotype.
- Automatic perception-behavior links may facilitate empathic reactions during social interactions.

Limitations Stated by Authors

- The limited range of behavioral options in the experimental situations may have inflated effect sizes.
- The experiments used specific situational contexts, limiting generalizability to other real-world settings.

Future Directions Proposed by Authors

- Investigate the boundary conditions under which automatic behavior can be controlled, focusing on awareness, motivation, and attentional capacity.
- Examine how pre-existing behavioral representations interact with situational cues in more diverse contexts.

Citations Referenced

- Lewin (1943) --
- Mischel (1973) --
- Berkowitz (1984) --
- Higgins (1987) --
- Carver et al. (1983) --
- Devine (1989) --
- Srull & Wyer (1979) --
- Bargh (1989) --
- Wegner (1994) --
- Bem (1972) --
- Bargh (in press) --
- Kihlstrom, Barnhardt, & Tatarzyn (1992) --
- Moore (1982) --

Critique

Paper:

bargh-et-al-2015-automaticity-of-social-behavior--direct-effects-of-trait-construct-and-stereotype-activation-on-a

Overall Assessment

The paper suffers from several moderate methodological and statistical shortcomings that undermine confidence in its central claim of direct, automatic trait-behavior links. The experimental designs conflate behavioral outcomes with perceptual judgments, employ weakly validated priming manipulations, and lack essential reporting of power analyses, exclusion criteria, and reliability metrics, while key statistical tests are missing. Consequently, the evidentiary support for the purported automatic effects is tenuous, and the theoretical framing is not convincingly anchored to the observed data.

1. Design-Theory Alignment

- **The interruption waiting time outcome conflates participants' behavioral response with their perception of the experimenter's behavior, making it unclear whether the effect reflects a direct trait-behavior link or a mediated perceptual judgment.**

- Severity: **moderate**

• Justification: The theoretical claim is that primed trait concepts directly trigger corresponding behavior. However, the measured latency to interrupt could be driven by participants' appraisal of the experimenter's rudeness, which is a perceptual judgment rather than a pure behavioral enactment of the primed trait. This conflation weakens the alignment between the predictor (trait priming) and the intended outcome (automatic behavior).

Evidence:

- `results_and_discussion/exp_1` -- field `key_findings`: Authors note a possible alternative explanation that rude-primed participants may have perceived the experimenter as rude, influencing their decision to interrupt.
- `results_and_discussion/exp_1` -- field `alternative_interpretations_discussed`: Discussion of how participants might have interpreted the experimenter's ambiguous behavior differently depending on priming.

The scrambled-sentence priming materials contain many filler words unrelated to the target trait, so the manipulation may not isolate activation of the specific trait construct (rudeness or politeness) that the theory predicts.

- Severity: **minor**

• Justification: If only a subset of the presented words actually cues the trait, the manipulation may only partially activate the theoretical construct, limiting confidence that any observed behavioral change is due to the intended trait activation rather than incidental processing of unrelated content.

Evidence:

- methods/exp_1 -- field materials: Three versions differ in 15 critical words per version, but the remaining 15 words are unrelated filler items.

In Experiment 2, walking speed is used as a behavioral proxy for the elderly stereotype, yet the stereotype encompasses multiple attributes (e.g., frailty, slowness, negativity) and the outcome captures only one facet, potentially misrepresenting the theoretical construct.

- Severity: **moderate**
- Justification: The elderly stereotype includes a range of behavioral expectations (e.g., reduced mobility, slower speech, increased deference). Measuring only walking speed may not capture the full behavioral representation the theory claims to activate, limiting the test's fidelity to the theoretical mechanism.

Evidence:

- methods/exp_2a_and_2b -- field outcome_measures: Outcome is time to walk 9.75 m, operationalized as walking speed.
- introduction -- field theoretical_framework: Theory posits that activated behavioral representations linked to a stereotype should be triggered automatically.

Experiment 3's hostility rating relies on coders' judgments of facial expressions after a computer error, which may reflect participants' emotional response to the error rather than a direct behavioral enactment of the African-American stereotype.

- Severity: **moderate**
- Justification: The theoretical claim is that stereotype activation directly produces hostile behavior. However, the measured outcome is an observer-rated facial expression following an unrelated task error, which could be driven by frustration or task difficulty rather than the specific stereotype-driven hostility the theory predicts.
- Evidence:
- methods/exp_3 -- field outcome_measures: Hostility rating is a 5-point scale coded from videotaped facial expressions.
- results_and_discussion/exp_3 -- field authors_interpretations: Authors claim the effect is nonconscious, but alternative explanation of mood mediation is discussed.

2. Evidence-Theory Alignment

- **The authors claim that priming directly triggers behavior without mediation, yet the large ceiling effect and non-normal distribution in Experiment 1 undermine**

confidence that the observed faster interruptions reflect a direct automatic effect.

- Severity: **moderate**
- Justification: The theoretical claim that priming produces a direct, automatic behavioral response rests on a measure that suffered from a pronounced ceiling effect, making it unclear whether the effect is genuine or an artifact of measurement constraints. This weakens the alignment between the evidence and the claim of a direct, non-mediated influence.

Evidence:

- results_and_discussion/exp_1 -- field anomalous_or_unexpected_results: 21 of 34 participants did not interrupt within the 10-minute limit, creating a severe ceiling effect and non-normal distribution
- results_and_discussion/exp_1 -- field authors_interpretations: Authors interpret the faster interruptions as a direct preconscious effect despite acknowledging the distribution problem

In Experiment 2 the authors assert a nonconscious effect of elderly-stereotype priming on walking speed, yet they also report a significant mood difference that could plausibly mediate the behavioral outcome, which they downplay without sufficient evidence.

- Severity: **moderate**
- Justification: The claim that the walking-speed effect is purely nonconscious conflicts with the reported mood differences; dismissing mood as a mediator without further analysis (e.g., mediation testing) overstates the evidence and stretches the theoretical interpretation.

Evidence:

- results_and_discussion/exp_2a_and_2b -- field key_findings: Participants in the elderly prime condition reported a more positive mood ($M = 1.7$ vs 0.3) and higher arousal, yet the MANOVA showed no main effect for emotion/arousal
- results_and_discussion/exp_2a_and_2b -- field authors_interpretations: Authors claim the mood mediation explanation is untenable despite the observed mood differences

The general discussion extrapolates the findings to a broad theory of automatic behavior across diverse contexts, but the three experiments each involve highly constrained laboratory tasks with limited behavioral repertoires, which the authors acknowledge as a limitation.

- Severity: **minor**

- Justification: While the authors acknowledge limited behavioral options, they still present the results as evidence for a universal automatic perception-behavior link, which extends beyond the narrow set of measured behaviors and may mislead readers about the scope of the empirical support.
- Evidence:
- `general_discussion -- field limitations_stated_by_authors`: The limited range of behavioral options in the experimental situations may have inflated effect sizes and limits generalizability
- `general_discussion -- field summary_of_findings`: Authors claim behavior can be directly linked to situational representations and that effects are stronger than impression effects

5. Statistical Completeness

- **Experiment 2 did not report the expected interaction between prime condition and experiment (2a vs 2b) nor the main effect of experiment, despite a 2×2 between-subjects design.**
- Severity: **moderate**
- Justification: A factorial design with two factors requires reporting of each main effect and the interaction. The authors present only within-experiment t-tests, leaving the interaction and the experiment main effect unreported, which limits assessment of whether the priming effect is consistent across the two experiments.

Evidence:

- `methods/exp_2a_and_2b -- field design_statement`: 2 (prime condition: elderly vs neutral) $\times 2$ (experiment: 2a vs 2b) between-subjects design
- `results_and_discussion/exp_2a_and_2b -- field key_findings`: Only separate t-tests for each experiment are reported; no ANOVA interaction term or main effect of experiment is presented

Experiment 3 lacks any reported statistical analysis of the hypothesized effect of African American versus Caucasian priming on hostility ratings.

- Severity: **moderate**
- Justification: Given a simple two-level between-subjects design, the authors should report at least a t-test or ANOVA comparing hostility ratings across the two priming conditions. The absence of any statistical report prevents verification of the claimed effect.

- Evidence:
- `methods/exp_3 -- field design_statement: 1` (priming: African American vs Caucasian) between-subjects design
- `results_and_discussion/exp_3 -- field key_findings: No results or statistical test entries are provided for Experiment 3`

6. Methodological Concerns

- **No power analysis reported for any of the three experiments, leaving the adequacy of sample sizes unclear.**
- Severity: **moderate**
- Justification: The methods sections for Experiments 1, 2 (both 2a and 2b), and 3 all state that a power analysis was not reported. Without a priori power calculations, it is uncertain whether the sample sizes (34, 60, and 41 participants respectively) provide sufficient statistical power to detect the hypothesized effects, especially given the small per-condition Ns.

Evidence:

- `methods/exp_1 -- field power_analysis: null`
- `methods/exp_2a_and_2b -- field power_analysis: null`
- `methods/exp_3 -- field power_analysis: null`

Exclusion criteria are not specified in any experiment, making it unclear how data were screened or participants removed.

- Severity: **moderate**
- Justification: All three experiments list empty exclusion arrays and note in the "missing_or_unclear" fields that exclusion criteria were not reported. Transparent reporting of exclusion rules is essential for reproducibility and for assessing potential biases in the final sample.

Evidence:

- `methods/exp_1 -- field exclusions: []` (no criteria described)
- `methods/exp_2a_and_2b -- field exclusions: []` (no criteria described)
- `methods/exp_3 -- field exclusions: []` (no criteria described)

Counterbalancing procedures are not described despite designs that could benefit from it (e.g., order of stimulus presentation, assignment of prime versions).

- Severity: **minor**
- Justification: Each methods section includes a "counterbalancing" entry marked as "not reported". For experiments involving multiple stimulus versions (e.g., different scrambled-sentence lists) or trial orders, lack of a clear counterbalancing scheme could introduce order effects or confounds.

Evidence:

- methods/exp_1 -- field counterbalancing: not reported
- methods/exp_2a_and_2b -- field counterbalancing: not reported
- methods/exp_3 -- field counterbalancing: not reported

Reliability of the coded hostility rating in Experiment 3 is not reported, despite using two blind coders.

- Severity: **moderate**
- Justification: The outcome measure relies on subjective coding of facial expressions. Without reporting inter-rater reliability (e.g., ICC or Cohen's κ), the consistency of the hostility scores cannot be assessed, which threatens the validity of the dependent variable.

Evidence:

- methods/exp_3 -- field outcome_measures: hostility rating coded by two blind coders, but no inter-rater reliability statistic provided

Random assignment procedures are not detailed; it is unclear how participants were allocated to priming conditions.

- Severity: **minor**
- Justification: While the procedure notes computer-based random assignment for Experiment 3, the lack of detail about the randomization method (e.g., block randomization, seed) makes it difficult to verify that allocation was truly random and balanced across conditions.
- Evidence:
 - methods/exp_3 -- field procedure: participants randomly assigned to priming condition by computer, but no description of randomization algorithm or checks
 - methods/exp_2a_and_2b -- field design: prime condition assigned between participants, but counterbalancing not reported